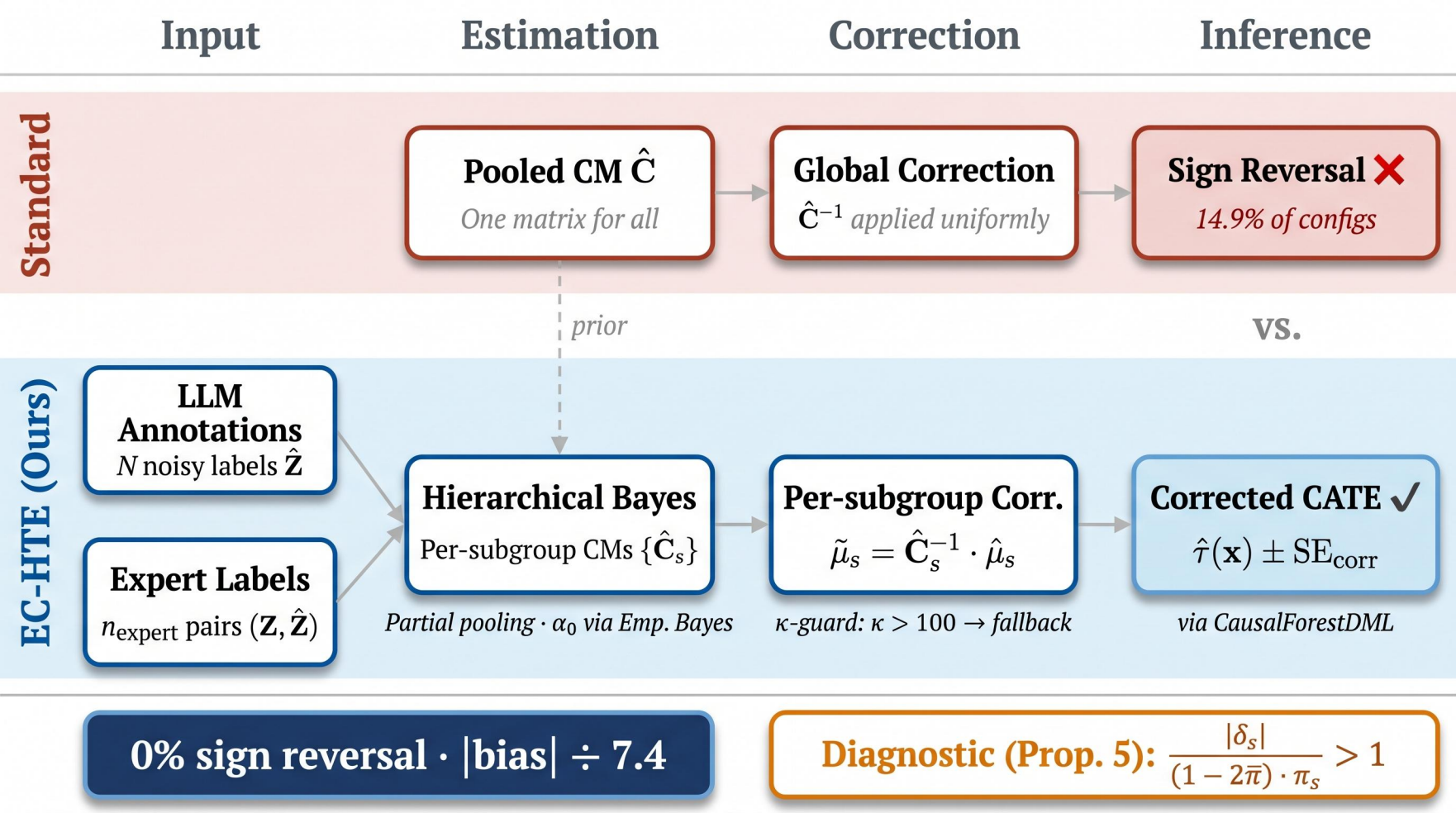

Input
Estimation
Correction
Inference
Standard
Pooled CM $\hat{C}$
One matrix for all
Global Correction
$\hat{C}^{-1}$ applied uniformly
Sign Reversal ✗
14.9% of configs
prior
vs.
EC-HTE (Ours)
LLM Annotations
N noisy labels $\hat{Z}$
Expert Labels
n_{expert} pairs $(Z, \hat{Z})$
Hierarchical Bayes
Per-subgroup CMs $\{\hat{C}_s\}$
Partial pooling · α_0 via Emp. Bayes
Per-subgroup Corr.
$\tilde{\mu}_s = \hat{C}_s^{-1} \cdot \hat{\mu}_s$
κ-guard: $\kappa > 100 \rightarrow$ fallback
Corrected CATE ✓
$\hat{\tau}(x) \pm SE_{corr}$
via CausalForestDML
0% sign reversal · |bias| ÷ 7.4
Diagnostic (Prop. 5): $\frac{|\delta_s|}{(1-2\bar{\pi}) \cdot \pi_s} > 1$